

Abstract

1. Abstracts explain to the general reader why the research was done and why the results are important. They should start with some brief background information: a sentence giving a **broad introduction** to the field comprehensible to the general reader, and then a sentence of **more detailed background specific to the author's study**. This should be followed by the **results**, or if the paper is more methods/technique-oriented, an explanation of objectives/methods and then the results. The final sentence should outline the **main conclusions** of the study, in terms that will be comprehensible to all readers. Abstract length should be 125 words or less.
2. Include major terms and acronyms in the abstract, because the text of the abstract can be searched in many data-retrieval systems. This will allow the article to be retrieved when relevant.
3. If not specified in the title, indicate in the abstract the animal model or system in which the experiments were performed.
4. Do not cite references in the abstract. (The investigator's name may be introduced instead.)
5. Do not cite figures or tables in the abstract.
6. *P* values and other statistical information may appear in the abstract at the editor's discretion.
7. Spell out all units of measure (including kelvin), except M (molar concentration), ° (e.g., latitude or angle), °C, and %.
8. Spell out the names of amino acids (e.g., leucine-11) and use abbreviations for nucleotides (C, G, A, T).
9. Define cDNA, mRNA, and tRNA the first time they are used in the abstract.

Acronyms

1. Spell out in the abstract all acronyms used in the title.
2. Use abbreviations and acronyms in the abstract only when it is necessary to avoid awkward construction or needless repetition. For example, the unit kilograms per cubic meter (kg m^{-3}) may be defined on first usage and the abbreviation used thereafter.
3. Acronyms introduced in the abstract should be redefined at first use in the text.

Elements and compounds

1. Spell out chemical elements (carbon, oxygen, etc.).
2. Spell out all chemical names and formulas where possible; for example, avoid NaCl (sodium chloride), GABA (γ -aminobutyric acid), HCO_3^- (bicarbonate ion).
3. Indicate the ionic state of a compound when necessary; for example, calcium ions (Ca^{2+}).
4. Isotopes and isotope ratios: When an isotope is mentioned more than once, spell out the name of the element and then give the isotope number; for example, carbon-12 (^{12}C), sodium-23 (^{23}Na). When isotope ratios are presented, abbreviations may be used; for example, $^{14}\text{C}/^{12}\text{C}$, $^{40}\text{K}/^{40}\text{Ar}$.

Grammatical tense

The simple past tense should be used to state the completed operations, observations, and data of the experiments reported in the paper. The present tense should be used for established facts, previous findings, and conclusions of the study. Note: Authors often discuss particular events and general truths together, and so they may switch back and forth between present and past tenses.

“First,” “new,” “novel,” etc.

1. Avoid any references to “first” to describe the present work. Avoid claims such as “Our findings are the first to demonstrate that...” or “Our laboratory has pioneered...”.
2. Avoid using terms such as “novel,” “new,” or “unique” to characterize the findings presented in a paper, unless a truly new thing is being discussed (such as a new type of instrument). Such terms can often simply be deleted (see example 1 below) or rephrased (examples 2 and 3). Genes and proteins are almost never “new” and should instead be described as “previously unidentified/unknown/uncharacterized/unrecognized.” When describing an unanticipated or surprising role for a known gene or protein, “unexpected” or “unusual” may be used in place of “novel.”

Example 1.

“The determination of the chimpanzee genome sequence provides a **new** means to study both structural and functional aspects of the evolution of the human genome.”

Comment: Here one can simply delete “new” without affecting the meaning.

Example 2. “Through a forward genetic screen, we identified a **novel** gene, *sleepless*, required for sleep in *Drosophila*.”

Comment: Delete “novel”; the gene is not new but previously unidentified.

Example 3.

“We describe a **new** DNA sequencing technology. A commonly available, inexpensive epifluorescence microscope can be converted to rapid nonelectrophoretic DNA sequencing automation.”

Comment: “New” is OK here if the authors are describing a new technology that they have invented.